

Masters Research Proposal

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Title: Solving Nonlinear PDEs via Quantum Singular Value Transformation

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1 Introduction

1.1 Aims

This thesis aims to develop and evaluate quantum algorithms for solving nonlinear partial differential equations (PDEs) using the Quantum Singular Value Transformation (QSVT) framework. Based on recent work that represents differential equations and their boundary conditions as effective Hamiltonians, this work extends the QSVT to the Generalised Quantum Singular Value Transformation (GQSVT). This extension enables the application of this method to complex-valued differential equations which arise naturally in many areas of physics and engineering. Additionally the project will explore how to incorporate data-informed modifications into these Hamiltonians, with the goal of modelling data-rich regimes.

1.2 Significance

Solving complex nonlinear partial differential equations (PDEs) on quantum hardware is a rapidly emerging prospect with broad applications ranging from fluid dynamics and turbulence modelling to quantum field theory. Established algorithms such as the Harrow-Hassidim-Lloyd (HHL) algorithm and variational methods typically rely on discretisation, which inhibits their scalability to stiff, nonlinear, multiscale or dense systems with fine grid sizes. In contrast, the effective Hamiltonian approach uses spectral methods to encode the solution into a Chebyshev polynomial basis, allowing it to surpass performance when fine-grid encoding is required. The quantum singular value transformation (QSVT) provides a flexible and modular approach to quantum linear algebra, allowing operations like matrix inversion under broader conditions than HHL [1]. However, the current method restricts effective Hamiltonians to real values, thereby limiting the class of operators and solutions to purely real outcomes. Moreover, the HHL paper only addresses linear partial differential equations or nonlinear ordinary differential equations

through a process called linearisation [3] and variational algorithms require classical optimisation [7].

This research is significant because it extends the effective Hamiltonian based framework to complex-valued nonlinear PDEs, which are common in frontier areas of various fields. In domains like geophysical inversion, deep turbulence and real-time weather prediction, incorporating data-informed terms into the Hamiltonian provides a promising path towards optimising quantum solutions to problems where complete physical models are unavailable or difficult to create. By enabling hybrid physics and data driven Hamiltonians, this work could expand the applicability of quantum PDE solvers to data-rich, real world problems.

2 Background

2.1 Problem Statement

Many physical systems are described by partial differential equations (PDEs), but solving them remains computationally costly. In the absence of closed-form solutions, many advanced classical algorithms implement numerical methods which are difficult to scale and lose accuracy in systems with a need for more fine grid or time-step discretisations. This is especially true of nonlinear PDEs which are the primary focus of this research.

Quantum algorithms provide a potentially scalable alternative. Quantum Singular Value Transformation (QSVT) allows quantum circuits to encode real-valued polynomials of definite parity. Recent theoretical developments have shown that it is possible to represent differential equations and their boundary conditions as effective Hamiltonians, which are structured matrices that can be processed using QSVT. However, due to QSVT's restriction to real-valued Hermitian Hamiltonians, the range of solvable PDEs is restricted. Systems with damping, gain/loss or complex boundary interactions can lead to complex-valued or non-Hermitian operators.

While novel frameworks such as the Generalised QSVT (GQSVT) have been proposed to handle complex-valued operators and remove parity constraints, they have not been systematically applied to nonlinear PDEs.

The existing effective Hamiltonian approach is largely physics-driven, assuming known governing models. Yet in many high-dimensional, data-rich domains such as geophysical imaging, real-time weather prediction, or turbulence modelling, the governing equations may be incomplete or unavailable. In these settings, data-informed Hamiltonian construction offers a promising path forward towards more flexible and accurate quantum solvers.

This research addresses the gap in current quantum PDE solving frameworks by developing and applying GQSVT based methods for nonlinear, complex-valued PDEs and exploring hybrid physics and data driven constructions for real world data-intensive applications.

2.2 Literature Review

2.2.1 Exploration of Problems

Burgers Equation (Viscous)

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2} \quad (1)$$

This equation appears in turbulence theory and has a closed form solution in one real spatial dimension with $\nu > 0$. However, complex versions do exist and the equation remains unsolvable analytically outside these circumstances. The nonlinearity term is $u \frac{\partial u}{\partial x}$.

Nonlinear Schrödinger Equation

$$i \frac{\partial \psi}{\partial t} + \nabla^2 \psi + |\psi|^2 \psi = 0 \quad (2)$$

This equation describes the evolution of complex wave functions in nonlinear media. Its nonlinearity term is $|\psi|^2 \psi$.

Complex Ginzburg-Landau Equation

$$\frac{\partial A}{\partial t} = A + (1 + i\alpha) \nabla^2 A - (1 + i\beta) |A|^2 A \quad (3)$$

This equation models phenomena such as superconductivity and fluid convection patterns. The nonlinearity term is $|A|^2 A$.

The current physics-informed Hamiltonian method is able to address Burgers equation in its real case, but the Schrödinger and Ginzburg-Landau equations involve are complex-valued fields. This limitation motivates the development of a more robust framework capable of handling a broader class of nonlinear PDEs.

2.2.2 Existing Classical and Quantum Solving Methods

Classical methods for solving PDEs employ mesh based discretisation techniques such as finite difference and finite element methods or spectral methods [11]. These methods approximate the continuous domain onto a discrete grid, converting the PDE into a large system of algebraic equations that can be solved numerically. These methods are well-established and reliable, but for high-resolution or multi-dimensional systems the computational cost increases rapidly. For higher dimensional problems, direct matrix or classical variational methods may be employed [4].

The Harrow-Hassidim-Lloyd (HHL) algorithm was one of the first to demonstrate exponential speedup for solving systems of linear equations, which arise from discretised PDEs [3]. HHL encodes the linear system $A\vec{x} = \vec{b}$ into a quantum state and uses phase estimation to prepare a quantum state proportional to the solution vector $\vec{x}$. However, it is limited to sparse, well-conditioned Hermitian matrices. This makes it impractical for solving PDE problems, particularly nonlinear ones. Importantly, in the 2021 "Grand Unification" paper, it is shown that HHL can be viewed as a special case of QSVT, where

we're looking to find a target polynomial that inverts the singular values of the matrix A .

Variational quantum algorithms use a hybrid quantum-classical loop to optimise a parameterised quantum circuit that approximates the solution to a PDE [7]. This approach also discretises the domains into a grid. While more flexible than HHL, these methods require classical optimisation and suffer from challenges such as barren plateaus and scaling issues for larger systems.

2.2.3 Quantum Singular Value Transformation

The quantum singular value transformation (QSVT) is a powerful framework that generalises quantum linear algebra operations by enabling quantum circuits to apply polynomial transformations to the singular values of block encoded matrices [1]. QSVT enables the application of any parity-constrained polynomial $f(\Sigma)$ to the singular values Σ of a matrix A block encoded within a unitary.

$$U = \begin{bmatrix} A & \cdot \\ \cdot & \cdot \end{bmatrix} \text{ with } \|A\| \leq 1 \quad (4)$$

Assuming we have access to such a unitary U and the ability to apply a sequence of controlled-phase rotations by an angle ϕ , we can construct a unitary $V_{\vec{\phi}}$, dependent on the values of $\vec{\phi}$ we select for our phase rotations, that looks like the following:

$$V_{\vec{\phi}} = \begin{cases} \tilde{\Pi}_{\phi_1} U \left[\prod_{k=1}^{(d-1)/2} \Pi_{\phi_{2k}} U^\dagger \tilde{\Pi}_{\phi_{2k+1}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for odd } \text{Poly}^{(\text{SV})}(A) \\ \left[\prod_{k=1}^{d/2} \Pi_{\phi_{2k-1}} U^\dagger \tilde{\Pi}_{\phi_{2k}} U \right] = \begin{bmatrix} \text{Poly}^{(\text{SV})}(A) & \cdot \\ \cdot & \cdot \end{bmatrix} & \text{for even } \text{Poly}^{(\text{SV})}(A) \end{cases} \quad (5)$$

where the sequences of phases $\vec{\phi} = (\phi_0, \phi_1, \dots, \phi_d)$ is chosen specifically to encode a desired polynomial $f(x)$ of degree d [1]. $\tilde{\Pi}_\phi$ and Π_ϕ represent a controlled phase rotation by ϕ where $\tilde{\Pi}$ and Π are the left and right unitary projectors respectively such that $A := \tilde{\Pi} U \Pi$. By carefully selecting which phase angles to rotate by, we can approximate any real, fixed parity continuous polynomial [6] acting on the singular values of A .

It can be shown that QSVT serves as a unifying framework for many quantum algorithms including quantum search, Hamiltonian simulation, factoring, quantum walks and more [8]. A key limitation of QSVT is that it only supports real-valued polynomials applied to matrices. This restricts its applicability to problems where both the matrix and the function being applied are real-valued [1]. For differential equations, this implies only PDEs with real-valued operators and real-valued solutions can be handled.

For physics-informed partial differential equation (PDE) solvers, QSVT is used in ground-state preparation via Hamiltonian simulation. In this case the target polynomial is $p(x) = e^{-tx}$ [11].

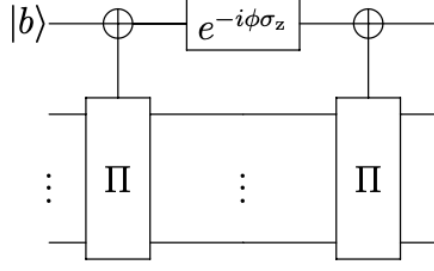


Figure 1: Quantum Circuit Diagram of the implementation of a Projector Controlled Phase Gate, Π_ϕ [2]

2.2.4 Physics-Informed Effective Hamiltonians

This algorithm encodes a differential equation in a latent Chebyshev basis and finds its solution by preparing the ground state of a specifically constructed Hamiltonian [11]. First, consider some differential equation with a differential operator $\mathcal{L}$, solution $u(x)$ and source function $f(x)$:

$$\mathcal{L}[u(x)] = f(x), \quad x \in [a, b]. \quad (6)$$

We can represent the solution as a state overlap in the Chebyshev basis with a scale factor s , namely

$$u(x) = s \langle T(x) | \psi \rangle \quad (7)$$

Here $|T(x)\rangle$ is known as a "feature map" state, a superposition of computational basis states with Chebyshev coefficients. $|\psi\rangle$ is a parameterised quantum state carrying the unknown amplitudes. Thus, we've effectively represented our solution in a latent Chebyshev basis [11].

To translate differential operators, we use the known differentiation matrices associated with the Chebyshev basis [5]. An example translation is shown below where $\mathbb{G}$ is the generator matrix corresponding to differentiation:

$$\frac{d^2 f(x)}{dx^2} + a f(x) = 0 \longrightarrow \mathbb{A} = \mathbb{G}^T{}^2 + a \mathbb{I}. \quad (8)$$

Here, $\mathbb{A}$ is the matrix corresponding to the latent space representation of the DE. Crucially, differential operators act simply in the Chebyshev basis, as the following

$$\frac{d}{dx} f(x) = \frac{d}{dx} \langle T(x) | \psi \rangle = \langle T(x) | D^\dagger | \psi \rangle. \quad (9)$$

As a result, all terms in the DE can be rewritten as linear operators acting on $|\psi\rangle$.

To implement constraints, we use $\mathbb{B}(x)$, a data transformation operator dependent on x . For example, for specific constraints such as $f(x_0) = 0$ or $f'(x_1) = 0$, we can use $\mathbb{B}(x_0)$ or $\mathbb{B}(x_1)\mathbb{G}^\dagger$ [11]. From these, we can create our effective Hamiltonian:

$$\mathbb{H} = \mathbb{A}^\top \mathbb{A} + \mathbb{B}^\top \mathbb{B}. \quad (10)$$

We may add additional boundary constraints as required, which would effectively be more $\mathbb{B}$ matrices. Therefore, by construction, solving the original DE is equivalent to solving the eigen-problem:

$$\mathbb{H}_{\text{eff}}|\psi\rangle = 0, \quad (11)$$

and the desired solution corresponds to the ground state of the effective Hamiltonian. $\mathbb{H}$ is a real symmetric positive-definite matrix with distinct eigenvalues and mutually orthonormal eigenvectors [11].

We then block encode the effective Hamiltonian, and use a ground state preparation routine, QSVT, to amplify the amplitude of the ground state. Specifically, we use quantum imaginary time evolution, constructing the polynomial approximation of $e^{-t\mathbb{H}}$. This exponentially suppresses the excited eigen-components, leaving a higher probability of measuring the ground state. Importantly, the algorithm directly prepares the lowest energy solution without classical optimisation [11].

Finally, to measure the quantum state, we use a Chebyshev transformation and Hadamard test to measure the state we have prepared. This controlled overlap test effectively measures the overlap $\langle T(x)|\psi\rangle$. In practice this is done via interferometric measurement [11].

This same framework is extendable to multi-variable and nonlinear DEs. For a function of two variables $f(x, y)$, we use a tensor product Chebyshev basis with two registers, preparing a joint feature state $|T(x)\rangle \otimes |T(y)\rangle$. Take Laplace's equation for example:

$$\frac{d^2 f}{dx^2} + \frac{d^2 f}{dy^2} = 0 \longrightarrow \mathbb{A} = \mathbb{G}_x^{\top 2} \otimes \mathbb{G}_y^{\top 2}. \quad (12)$$

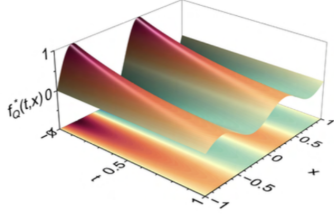
For nonlinear terms, we add extra registers. For example, a term like $f'(x)^2$ leads to a product of two copies of $|\psi\rangle$ in latent space. We create a new, higher dimensional latent Chebyshev space where $|\Psi\rangle \equiv |\psi\rangle \otimes |\psi\rangle$, so that the nonlinearity term becomes linear in the larger space. This naturally handles all nonlinearity. Inhomogeneous terms are handled via Maclaurin series expansion. For all these cases, we then construct the effective Hamiltonian as usual and solve [11]. This approach allows us to encode the solution over the entire domain, rather than using grid discretisation. Therefore, it is possible to evaluate the solution function at any arbitrary input values. However, as discussed above, QSVT can only handle real values. This restricts the class of differential equations we're able to solve.

Example Solutions:

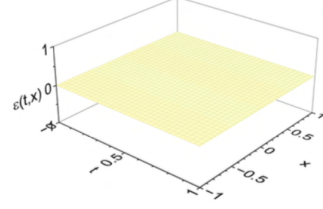
Below are results from the application of this algorithm on example differential equations [11]. The paper did not attempt to solve nonlinear PDEs, but there are examples of both linear PDEs and nonlinear DEs. $f(x)$ is the known solution and $f_Q^*(x)$ is the quantum solution.

1. Heat Equation

$$\frac{\partial f(t, x)}{\partial t} - k \frac{\partial^2 f(t, x)}{\partial x^2} = 0, k = \frac{1}{25}, \mathcal{BC} = \{f(0, x) = \sin(2\pi x), f(t, \pm 1) = 0\} \quad (13)$$



Solution



Error

Figure 2: Solution for the Heat Equation Example. [11]

2. Nonlinear Example 1

$$4 \frac{d^2 f(x)}{dx^2} + 2 \left(\frac{df(x)}{dx} \right)^2 + f(x) = 0, \mathcal{BC} = \{f(0) = 1, f'(0) = 0\} \quad (14)$$

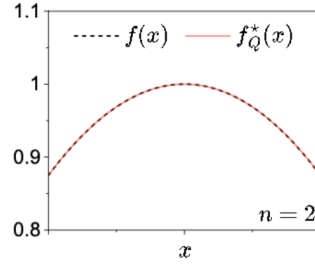


Figure 3: Solution of Nonlinear Example 1 [11]

3. Nonlinear Example 2

$$\frac{d^2 f(x)}{f x^2} - 2f(x)^2 + x = 0, \mathcal{BC} = \{f(-1) = -0.1, f(1) = 0.1\} \quad (15)$$

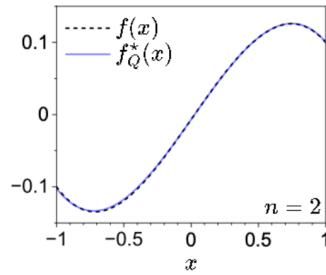


Figure 4: Solution of Nonlinear Example 2 [11]

The model performs well for all examples shown. The errors are almost negligible and decrease as the number of qubits used (shown as n on the figures) increase. The experiments were conducted on classical simulations of quantum hardware and are therefore currently computationally costly, however based on these results it is projected this algorithm will become useful in the early fault-tolerant era [11].

2.2.5 Generalised Quantum Singular Value Transformation

The generalised QSVT (GQSVT) framework extends the standard QSVT algorithm by lifting the restrictions on coefficient phases and polynomial parity [9]. The standard QSVT restricts the target polynomial $p(x)$ to have real coefficients and definite parity. However, the generalised version allows complex polynomials and indefinite parity. These generalisations are inherited from developments in generalised quantum signal processing (GQSP) which is itself an extension of the quantum signal processing framework that QSVT was built on [10]. Eliminating the restrictions on real polynomials and Hamiltonians imposed by QSVT will allow us to solve complex-valued differential equations. GQSVT also promises a simplification in complexity for the crucial and time-consuming step of calculating phase factors. The task was of complexity $\mathcal{O}(d^2)$ for QSVT, GQSVT promises a complexity of $\mathcal{O}(d)$ [9]. This should speed up calculations even for problems which are currently possible via the existing method.

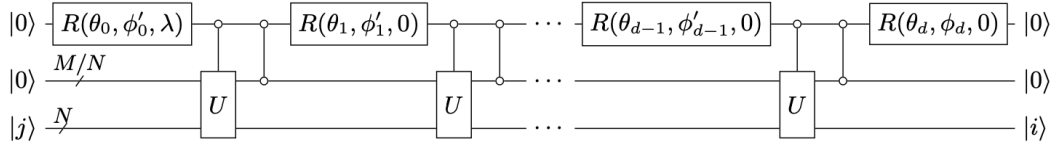


Figure 5: Example circuit for the generalised quantum singular value transformation for a Hermitian matrix A , block encoded as U . R is a rotation matrix analogous to Π for QSVT [9].

3 Research Plan

3.1 Method

This research will be primarily theoretical and computational. The goal is to design, simulate and evaluate novel quantum algorithms for solving nonlinear, complex-valued partial differential equations (PDEs). Development will be conducted using using *Mathematica* for symbolic manipulation and proof-of-concept derivations. Simulations will be created using *Python* based simulators such as *Qiskit* and *PennyLane* and results will be compared to classical solving algorithms. Efforts will be split into two phases. First, extending the current QSVT algorithm to GQSVT and second, integrating data-informed terms into effective Hamiltonians to enable solving PDEs in data-rich regimes. This research represents a new research direction within QUISA.

Proposed Plan:

1. Reproduce the Current Physics-Informed QSVT Method:
 - (a) Implement the standard QSVT framework in *Python*.
 - (b) Choose an appropriate problem.
 - (c) Implement a physics-informed Hamiltonian PDE solver in *Python*.
 - (d) Compare quantum solution against classical benchmarks
2. Extend to GQSVT:
 - (a) Create the GQSVT algorithm in *Python*.
 - (b) Choose an appropriate problem.
 - (c) Implement a new physics-informed Hamiltonian PDE solver in *Python*.
 - (d) Compare quantum solution against classical benchmarks
3. Experiment with more data-informed terms in the Hamiltonian

Timeline:

Semester 1 2025: Research Proposal.

Semester 2 2025: Recreation of QSVT Physics-informed method. Begin extension to GQSVT.

Semester 1 2026: Completion of a working GQSVT enhanced algorithm, including simulation and evaluation in comparison to classical algorithms.

Semester 2 2026: Experimentation with data-informed terms.

3.2 Preliminary Work

In semester 1 2025, work consisted of a thorough literature review, achieving an in-depth understanding of QSVT and the physics-informed Hamiltonian approach for PDEs. This was pivotal in developing a thesis that could both build on the newest developments in the field and be achievable within the time-frame of a Master's course. A working *Python* implementation of QSVT has been created and work is underway on implementing the current physics-informed Hamiltonian approach using QSVT.

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